Damped Normal Modes

Consider 3 equal masses in a line held between two walls and 4 equal springs. A single small dashpot (C = $0.1\sqrt{mk}$) connects the leftmost mass from the wall at its left.

- Assume that there is no initial velocity
- Assume that the initial position is a mode shape with mass 1 having a displacement of 1
- Plot position vs time of mass 1 for all three mode shapes

Numerical ODE Solution

The Equation of motion for this system is:

$$\vec{M}\ddot{\vec{x}} + \vec{C}\dot{\vec{x}} + \vec{K}\vec{x} = \vec{F}(t) \tag{1}$$

Where:

$$\vec{M} = \begin{bmatrix} m1 & 0 & 0 \\ 0 & m2 & 0 \\ 0 & 0 & m3 \end{bmatrix} \tag{2}$$

$$\vec{C} = \begin{bmatrix} c1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{3}$$

$$\vec{K} = \begin{bmatrix} k1+k2 & -k2 & 0 \\ -k2 & k2+k3 & -k3 \\ 0 & -k3 & k3+k4 \end{bmatrix} \tag{4}$$

$$\ddot{\vec{x}} = \begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ \ddot{x}_3 \end{bmatrix} \tag{5}$$

$$\dot{\vec{x}} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} \tag{6}$$

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \tag{7}$$

$$\vec{F}(t) = F_o \begin{bmatrix} sin(\omega t) \\ 0 \\ 0 \end{bmatrix} \tag{8}$$

F_o is chosen to be equal to 3 for this problem. ω is the forcing frequency taken from the eigenvalues of K and M. This can be found using the **eig** command in MATLAB ([V,D] = eig(K,M)). The diagonals of D give the squares of modal frequencies of the each of the three modes present for the system,

$$\omega_1 = \sqrt{D(1,1)} \tag{9}$$

$$\omega_2 = \sqrt{D(2,2)} \tag{10}$$

$$\omega_3 = \sqrt{D(3,3)} \tag{11}$$

These can then be plugged into ODE45 to obtain a solution with the following command:

$$\mathrm{ode45}('\mathrm{DampedNormalModes\_RHS.m}', \mathrm{tspan}, \mathrm{z0}, \mathrm{options}, \mathrm{s})$$

where DampedNormalModes_RHS is the RHS file called to produce the numerical solution.

```
function zdot = DampedNormalModes_RHS(t,z,s)

x1 = z(1);
x2 = z(2);
x3 = z(3);

xdot1 = z(4);
xdot2 = z(5);
xdot3 = z(6);

x = [x1 x2 x3]';
xdot = [xdot1 xdot2 xdot3]';

M = s.M;
C = s.C;
K = s.K;
wm = s.wm;

F = [3*sin(wm*t); 0; 0];

xddot = -(M\C*xdot + M\K*x) + M\F;

xddot1 = xddot(1);
xddot2 = xddot(2);
xddot3 = xddot(3);

zdot = [xdot1 xdot2 xdot3 xddot1 xddot2 xddot3]';


end
```

Figure 1: RHS code to calculate the numerical solution

Analytic Solution

The Following MATLAB Code snippet shows how to obtain the analytic solution for one mode. This process can be repeated for the other two:

```
%% Analytic Method - 1st Normal Mode
%%Find the Particular Solution (Analytic)
wm = sqrt(D(1,1));

syms x t ct st real
syms a b p d e f real
x = [a;b;p]*sin(wm*t)+[d;e;f]*cos(wm*t);
xdot = jacobian(x,t);
xddot = jacobian(xdot,t);

eq_0 = M*xddot + C*xdot + K*x == [3*sin(wm*t) 0 0]';

eq_sub = subs(eq_0,[cos(wm*t), sin(wm*t)],[ct, st]);

eq_ct = jacobian(eq_sub,ct);
eq_st = jacobian(eq_sub,st);

[Amatrix,Bmatrix] = equationsToMatrix([eq_ct ;eq_st],[a b p d e f]);
Ad=double(Amatrix);
xhat = double(Amatrix\Bmatrix);


%%Particular Solution
t = linspace(0,tspan(2),1000);
x_p = [xhat(1); xhat(2); xhat(3)]*sin(wm*t)+[xhat(4);xhat(5);xhat(6)]*cos(wm*t);
```

Figure 2: Code for Particular Solution

```
%%Find the Homogeneous Solution (Analytic)
A=[zeros(3,3) eye(3); -M\K -M\C];

for i=1:length(t)
    x_h(:,i)=[x_0 xdot_0]...
        *expm(A.*t(i));
end

%%Full Analytical Solution
x_a = x_h(1:3,:) + x_p(1:3,:);



%%Plots
figure(4)
hold on
plot(t,x_a(1,:),'b','LineWidth',3)
plot(t,x_a(2,:),'r')
plot(t,x_a(3,:),'k')

legend('m1','m2','m3')
title('Analytic 1st Normal Mode')
title('1st Normal Mode')
xlabel('Time (s)')
ylabel('X Position (m)')
```

Figure 3: Code for Homogeneous and Combined Solution

Results

Running each of these through ode45 in MATLAB gives the following plots:

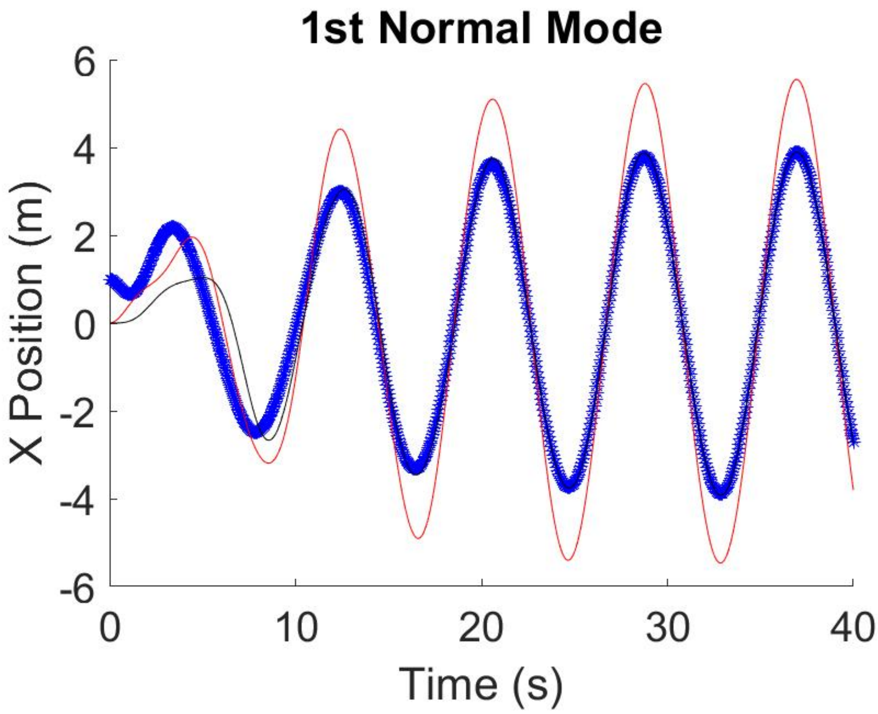

Figure 4: ODE45 plot of the mass positions for the first normal mode

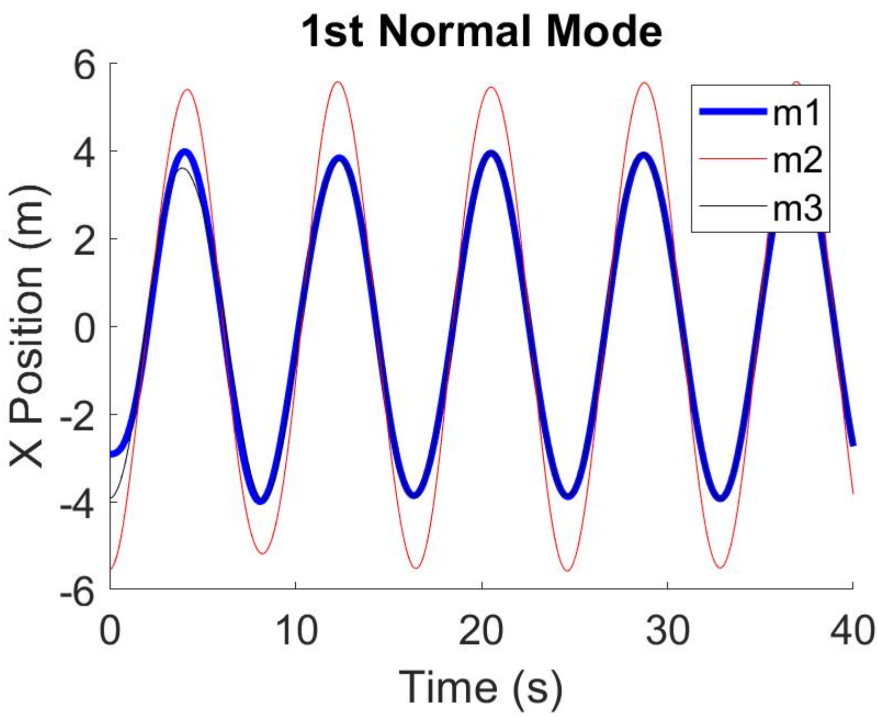

Figure 5: Analytical plot of the mass positions for the first normal mode

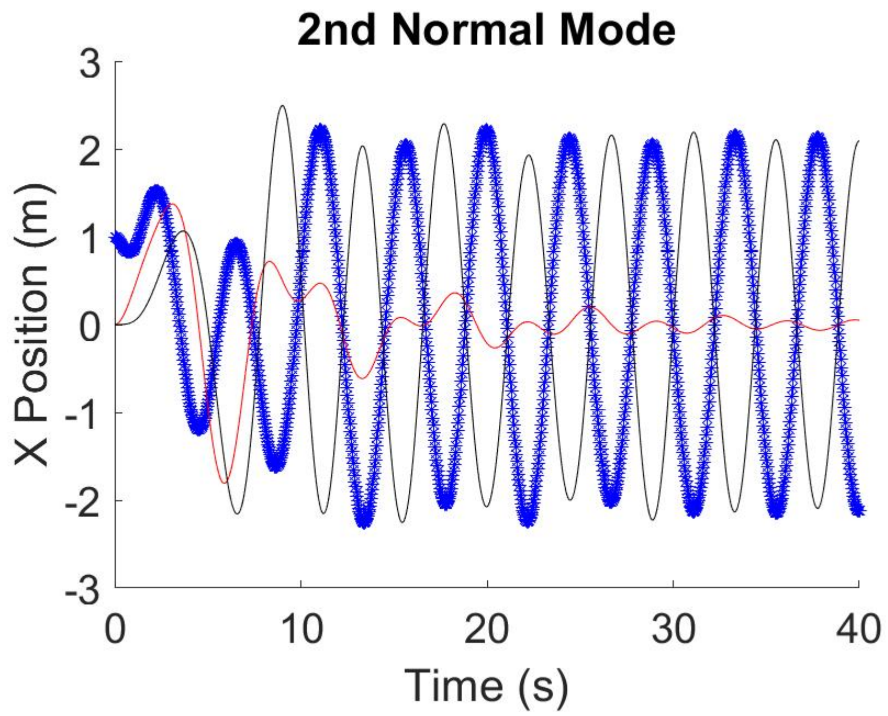

Figure 6: ODE45 plot of the mass positions for the second normal mode

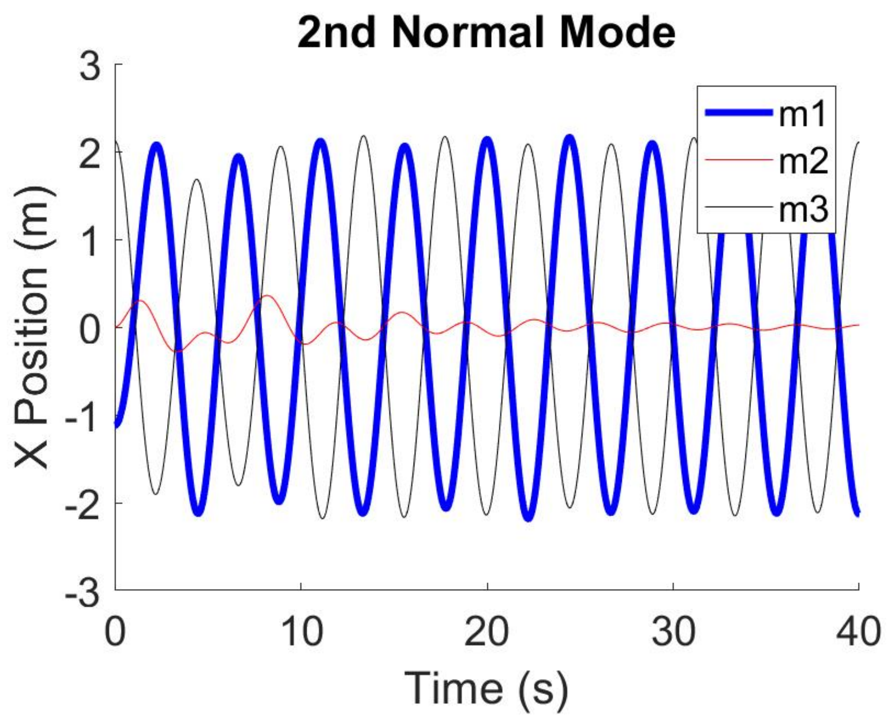

Figure 7: Analytical plot of the mass positions for the second normal mode

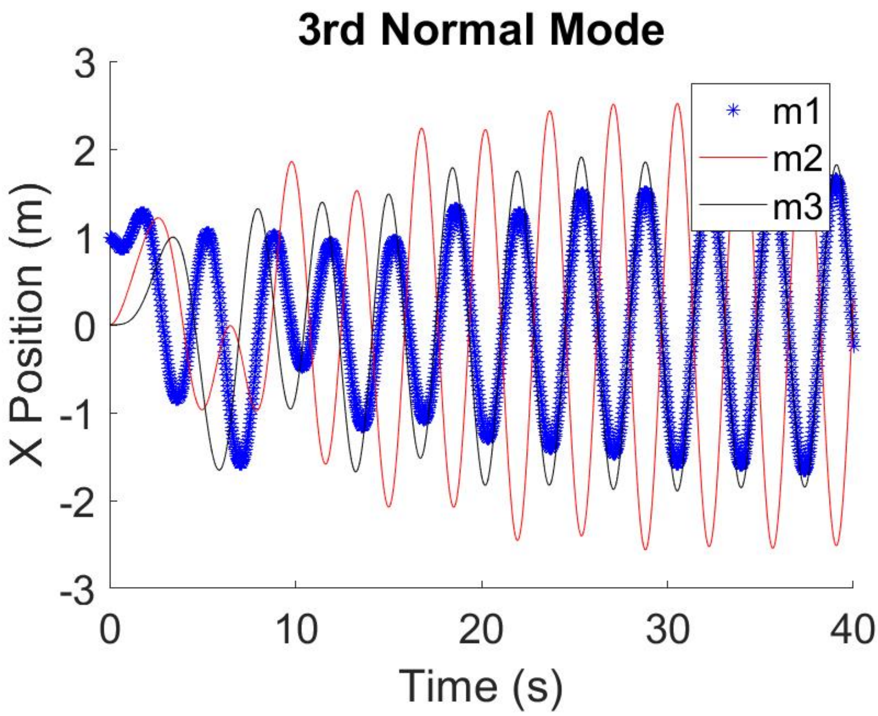

Figure 8: ODE45 plot of the mass positions for the third normal mode

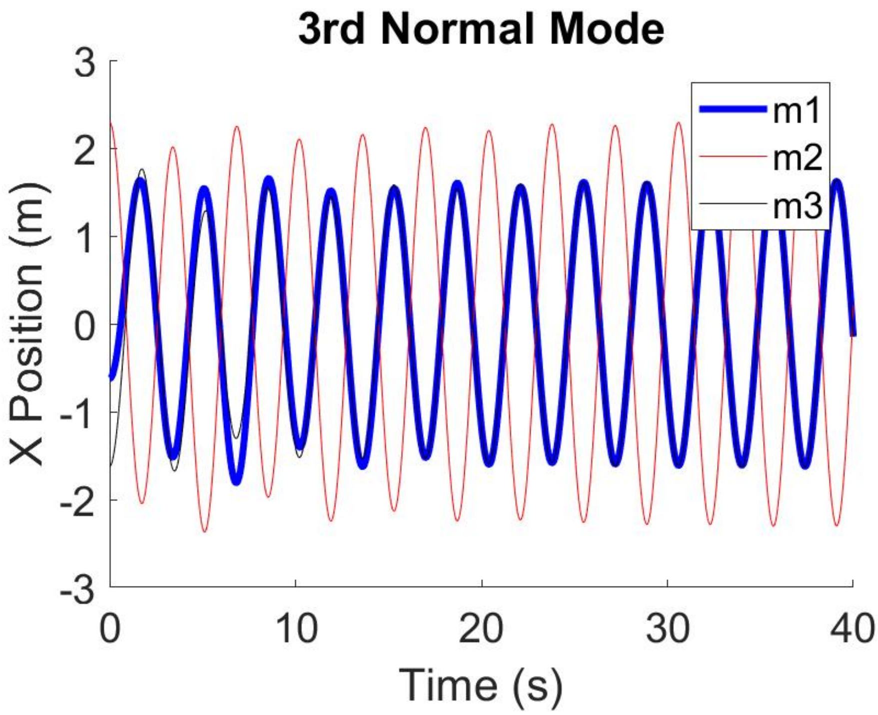

Figure 9: Analytical plot of the mass positions for the third normal mode

Table 1: Behavior of the Damped normal Modes

Mode	Steady State Behavior
1	Moves in Phase with the other two masses from its initial offset after some
2	m_1 and m_3 oscillate out of phase by 180 from each other, but m_2 remains nearly motionless, save for some small oscillations
3	m_1 and m_3 oscillate in phase, but m_2 oscillates 180 out of phase from the other two masses.

The Numerical solution appears to do a better job at prediction the particular motion than the analytical solution